Measurement Map

Companion to the manuscript skeleton, 10 October 2026

1 Scope

The map takes each object of the model in `MANUSCRIPT.tex` to its counterpart in the Basel capital framework [Basel Committee on Banking Supervision, 2011], the Basel liquidity standard [Basel Committee on Banking Supervision, 2013] and the Call Report instructions for regulatory capital [Federal Financial Institutions Examination Council, 2026], and says how it is observed. The FDIC panel fields are defined in the FDIC variable workbook [Federal Deposit Insurance Corporation, 2026]. The regulatory cap is that of Bayraktar et al. [2026] (BCVW). Quotation and reading IDs refer to the `lit/` records, where each quotation is checked verbatim on its page. The rearrangements of the regulatory ratios are proved in `lean/mathlib/MeasurementMap.lean`.

A status of Rule means a regulation sets the value. Report means a regulatory report records the object for each bank. Calibration means the model takes a value with no regulatory counterpart. Unobserved means neither a rule nor a report gives it. Output means the model produces it and its observation is part of the empirical design.

2 Terminology

One plain-English name for each object of the model. The manuscript and this map use these names and no others. Agreed means fixed with the author on 10 or 13 October 2026 (`notes/writing_discipline.md`, section 6). In use means the manuscript already uses the name consistently. Proposed means a plain name offered to the author and not yet agreed. The names in the last column are retired, and neither document uses them for these objects.

Table 1: Terminology.

Name	Symbol	Meaning	Status	Replaces
assets-to-liabilities ratio	x	Total assets over total liabilities, the state of the model.	Agreed	none
equity	$X - L$	Assets less liabilities.	In use	none
required capital	Q, q	The capital regulation requires, in dollars or as a ratio to liabilities.	Agreed	none
capital buffer	z, z_t	The log of equity over required capital. Zero when equity equals required capital, falling without limit towards the distress boundary.	Agreed	log coordinate
distress boundary	$x = 1$	The state at which equity is zero.	In use	deep in deficit
deficit	$z < 0$	Equity below required capital.	In use	none
surplus	$z > 0$	Equity above required capital.	In use	none
risky exposure	π	The share of assets held in the risky asset.	Agreed	none
regulatory cap	$u(x)$	The largest risky exposure regulation allows at each state, the smaller of its two branches.	In use	none
solvency branch	$(x-1)/a_1$	The branch of the regulatory cap set by capital. It binds near the distress boundary.	In use	none
liquidity branch	$(x-a_2)/a_3$	The branch of the regulatory cap set by liquidity. It binds deep in surplus.	In use	none
crossing point	$\bar{x}$	The state at which the two branches are equal.	In use	none

Name	Symbol	Meaning	Status	Replaces
cap geometry	a_1, a_2, a_3	The three parameters of the regulatory cap. Never a preference.	Agreed	none
asset volatility	σ	The volatility of the risky asset.	Agreed	none
expected asset return	μ_s	The drift of the risky asset.	Agreed	none
riskless rate	r	The return on the riskless asset.	In use	none
liability volatility	σ_L	The volatility of the shock to liabilities.	Agreed	none
return-liability correlation	c	The correlation between the risky return and the liability shock.	Agreed	none
net income on liabilities	γ	Income earned on liabilities, net of their cost.	In use	none
liability drift	μ_L	The drift of liabilities, $\gamma + r_L$.	In use	none
deposit rate	r_L	The rate paid on liabilities.	In use	none
discount rate	ρ	The shareholders' discount rate.	Agreed	none
liability-net discount rate	ρ_L	The discount rate of the problem written per unit of liabilities.	Agreed	none
return on equity	r_E	Net income over equity.	Agreed	none
dividends	D	Payments to shareholders, the singular control.	Agreed	none
dividend barrier	y^*	The state at which dividends are paid.	Agreed	payout barrier
recapitalisation	I	Capital raised in one issue, the impulse control.	Agreed	none
recapitalisation trigger	x_L	The state at which capital is raised.	Agreed	injection trigger, recapitalisation boundary
injection target	y_{post}	The state an issue reaches.	Agreed	none
fixed issuance cost	K	The cost of an issue that does not scale with its size, per unit of liabilities.	Agreed	none
proportional issuance cost	κ	The cost of an issue per unit raised.	Agreed	none
asymmetry parameter	λ_S	How much more the objective weighs a shortfall below the reference than a surplus above it.	Agreed	loss aversion
shortfall penalty	Λ	The term of the objective that charges a shortfall below the reference. Zero when the asymmetry parameter is one.	In use	none
reference	R	The level below which the shortfall penalty applies.	In use	none
expected discounted shortfall	W	The expected discounted shortfall below the reference under the optimal policy.	In use	none
discounted injection count	N	The expected discounted number of issues under the optimal policy.	In use	none
impulse operator	$\mathscr{M}V$	The value of issuing at once to the best target, net of both issuance costs.	In use	none
buffer variance	ζ	The variance per unit of time of the capital buffer.	Agreed	z-volatility, capital buffer's volatility
deficit limit	ν_1^2	The limit of buffer variance at the distress boundary.	Agreed	saturation limits
surplus limit	ν_3^2	The limit of buffer variance deep in surplus.	Agreed	saturation limits
variance ratio	λ_V^4, ν_1^2/ν_3^2	The deficit limit over the surplus limit.	Agreed	volatility ratio
induced variance asymmetry	λ_V	The asymmetry read off the variance ratio, distinct from the asymmetry parameter.	Agreed	none
saturation scale	θ	How sharply the variance of the discrete-time recursion moves between its deficit and surplus levels.	In use	none

Name	Symbol	Meaning	Status	Replaces
growth rate of required capital	g	The growth rate of required capital.	In use	none
retention ratio	b	The share of net income retained.	In use	none
surplus payout rate	δ_p	The share of surplus paid out in a period.	In use	none
deficit persistence	ϕ^-	How much of a deficit remains after one period.	In use	none
surplus persistence	ϕ^+	How much of a surplus remains after one period.	In use	none

3 Map

Table 2: Model objects and their observation.

ID	Object	Role in the model	Counterpart and observation	Status	Evidence
G1	x	The state, assets over liabilities.	Total assets over total liabilities, `ASSET/LIAB` in the FDIC panel. From the reported leverage ratio ℓ (Tier 1 over total assets for the leverage ratio), $x = 1/(1-\ell)$ holds only if Tier 1 equals book equity and the denominator equals total assets.	Report	`FDIC-Q1,FDIC-Q2,RCR-Q4,RCR-D3, MeasurementMap. leverage_of_state`
G2	π	The share of assets in the risky asset.	Risk-weighted assets over total assets, `RWAJT/ASSET`, under risk weights of 0 on the riskless and 1 on the risky asset. With a positive weight on the riskless asset the identification fails.	Report	`FDIC-Q3,RCR-Q5,RCR-D2,MeasurementMap. rwa_two_assets, MeasurementMap. control_rwa_needs_ zero_weight_on_ riskless`
G3	$X - L$	Equity, assets less liabilities.	CET1 capital or Tier 1 capital (`RBCT1J`) as reported. Both are net of regulatory adjustments, so they equal book equity only when the adjustments vanish.	Report	`FDIC-Q4,B3-Q3,RCR-Q1, RCR-Q3,B3-D2`
G4	a_1	The floor on equity over risky assets, the solvency branch $(x-1)/a_1$ of the cap.	The CET1 minimum of 4.5% of risk-weighted assets, which is BCVW's baseline 0.045, or the Tier 1 minimum of 6.0%. The bank's own ratio is reported as CET1 (`RBCT1CER`) or Tier 1 capital over risk-weighted assets. Banks electing the community bank leverage ratio report neither.	Rule	`FDIC-Q5,FDIC-Q6,BCVW-Q9,BCVW-Q13,B3-Q1,B3-Q2,B3-D1,RCR-Q2,RCR-Q6,RCR-Q7,RCR-D1, MeasurementMap. solvency_iff, MeasurementMap. control_solvency_ needs_positive_ exposure`
G5	a_2	Net 30-day outflows as a share of liabilities, in the liquidity branch $(x-a_2)/a_3$ of the cap.	A run-off rate on all liabilities. BCVW's baseline 0.05 is the factor for stable insured retail deposits, and less stable retail deposits run off at 10% or more. The LCR must be at least 100%. No report read gives the rate.	Rule	`BCVW-Q10,LCR-Q1,LCR-Q9,LCR-Q10,LCR-D1, MeasurementMap.lcr_ iff,MeasurementMap. control_lcr_needs_ positive_runoff`
G6	a_3	The haircut on the risky asset in high-quality liquid assets, the slope $1/a_3$ of the liquidity branch.	The riskless asset counts in full, as Level 1 assets do. The standard's haircuts are 15% (Level 2A) and 25% or 50% (Level 2B). BCVW's baseline 0.30 is none of them. The 40% cap on Level 2 assets and the 75% cap on inflows are not in the model.	Calibration	`BCVW-Q11,BCVW-Q12, BCVW-Q13,LCR-Q2,LCR-Q3,LCR-Q4,LCR-Q5,LCR-Q6,LCR-Q7,LCR-Q8,LCR-D2,LCR-D3, MeasurementMap.hqla_ haircut, MeasurementMap.cap_ iff`

ID	Object	Role in the model	Counterpart and observation	Status	Evidence
G7	q	The requirement on $x-1$, with $z=0$ at $x=1+q$.	A floor ℓ_0 on the leverage ratio is the floor $\ell_0/(1-\ell_0)$ on $x-1$, so the 3% Tier 1 leverage ratio gives $q=3/97$ when Tier 1 equals book equity. This is a reading, and the bundle does not fix q. Buffer variance does not depend on q.	Rule	`B3-Q7,RCR-Q4, MeasurementMap. leverage_req_iff, MeasurementMap. control_leverage_ needs_l0_lt_one, MeasurementMap.zvol2_ free_of_q`
G8	σ, μ_s, r	The volatility and drift of the risky asset and the riskless rate.	No regulatory counterpart. BCVW's baseline is $\sigma=0.08$, $\mu=0.04$, $r=0.02$.	Calibration	`BCVW-Q14`
G9	σ_L, c, γ, μ_L, r_L	The liability shock, its correlation with the risky return, the net income on liabilities, the liability drift and the deposit rate, with $\mu_L=\gamma+r_L$.	No regulatory counterpart. BCVW's baseline is $\sigma_L=0.03$, $c=0.20$, $\gamma=0.02$, $\mu_L=0.03$.	Calibration	`BCVW-Q14,M9`
G10	ρ_L	The liability-net discount rate of the reduced problem.	Formed from the shareholder discount rate, BCVW's baseline $\rho=0.12$. No regulatory counterpart.	Calibration	`BCVW-Q14`
G11	κ	The proportional issuance cost.	No regulatory counterpart. BCVW's baseline is $\kappa=0.01$ for dilution and $\kappa'=0.02$ per unit issued.	Calibration	`BCVW-Q14`
G12	K	The fixed issuance cost, charged per unit of liabilities.	Not observed. A cost fixed in dollars would break the reduction to x. Locally identified together with λ_S from the barrier and the trigger.	Unobserved	`BCVW-Q4,K2,M5,M6`
G13	λ_S	The asymmetry parameter of the shortfall penalty.	Not observed. The variance ratio carries no information about it. Locally identified together with K from the barrier and the trigger.	Unobserved	`K1,K2,M3,M6,M11`
G14	y^*	The dividend barrier.	The state at which the bank pays dividends (`EQCDIV`). Under the conservation buffer a bank with CET1 between 4.5% and 7.0% of risk-weighted assets has its distributions restricted, so observed payouts reflect the rule as well as the barrier.	Output	`FDIC-Q8,M4,B3-Q4,B3-Q5,B3-Q6,B3-D3`
G15	x_L, y_{post}	The recapitalisation trigger and the injection target.	The state just before and just after a recapitalisation, measured as net stock issuance (`EQCSXQ`) plus capital from the parent holding company (`EQCBHCTR`), in quarters without merger-driven equity changes (`EQCMRG`).	Output	`FDIC-Q9,FDIC-Q10, FDIC-Q12,FDIC-Q13,M5, M6,M7,M10`
G16	ζ, v_1^2, v_3^2	Buffer variance, with its deficit limit and its surplus limit.	The variance of changes in $z=\log((x-1)/q)$ near the distress boundary and deep in surplus. It does not depend on q.	Output	`M2,M3,M8, MeasurementMap.zvol2_ free_of_q`
G17	v_1^2/v_3^2	The deficit-to-surplus variance ratio, read in discrete time as λ_V^4.	The ratio of the two variances of G16. Reading it as λ_V^4 is a definition.	Output	`K1,M1,M3`
G18	z_t	The discrete-time state, the log of equity over required capital Q_t.	Tier 1 capital over 8.5% of risk-weighted assets, the Tier 1 minimum plus the conservation buffer.	Report	`FDIC-Q3,FDIC-Q4,B3-Q2,B3-Q4,M12`

ID	Object	Role in the model	Counterpart and observation	Status	Evidence
G19	g	The growth rate of required capital, exogenous in the model.	Quarterly growth of risk-weighted assets.	Report	FDIC-Q3,M12
G20	b	The retention ratio.	One minus dividends over net income, over four quarters, defined when net income is positive.	Report	FDIC-Q7,FDIC-Q8,M12
G21	δ_p	The surplus payout rate.	The share of the log-surplus paid as dividends in a quarter.	Report	FDIC-Q8,M12
G22	ϕ^-, ϕ^+	Deficit and surplus persistence of z_t.	Least squares of z_t on z_{t-1} through the origin, on pairs starting in deficit or surplus.	Output	M12

4 What the empirical analysis measured

Each object of the map, and what the FDIC panel of US banks, 2015 to 2024, could do with it. Tracked means the object is measured and no model prediction about it is tested. Holds and Fails mean a prediction of the solved model about it is tested. Not measured means the data available do not measure the object. Each figure is printed by the computation in the cited results file, which the checker re-runs on its pinned data.

Table 3: What the data could measure.

ID	Measured as	What the data show	Model prediction	Outcome	Evidence
G1	ASSET/LIAB, 213,771 bank-quarters.	Median 1.1182, interquartile range 1.0997 to 1.1447.	None on its own.	Tracked	FDIC-Q1,FDIC-Q2, 06_empirical_ starter/results_ model_and_data.md
G2	RWAJT/ASSET, under the risk-weight identification.	Median 0.6857.	$\pi = \bar{\pi}$, 2.3794 at the median state.	Fails	FDIC-Q3, MeasurementMap.rwa_ two_assets,06_ empirical_starter/ results_model_and_ data.md
G3	Book equity over assets, and Tier 1 capital (RBCT1J).	Book equity over assets has median 10.57%, interquartile range 9.06% to 12.64%.	None on its own.	Tracked	FDIC-Q4,06_ empirical_starter/ results_model_and_ data.md
G4	The bank's Tier 1 capital over risk-weighted assets.	Median 15.04%, 3.34 times a_1, and 2.78 times at the 25th percentile.	The ratio sits at a_1.	Fails	FDIC-Q4,B3-Q1,B3-D1,06_empirical_ starter/results_ model_and_data.md
G5	Not reported in the data read.	Nothing.	None tested.	Not measured	G5
G6	Not reported in the data read.	Nothing.	None tested.	Not measured	G6
G7	Not reported. The leverage ratio gives a candidate floor.	Nothing.	None tested.	Not measured	G7
G8	The volatility of book x is computed, which is not the economic σ.	Book volatility of x 0.01468 a year. The model's volatility of x is 0.06081 at the observed π, 4.1 times larger.	A level, on an economic-value basis.	Not measured	06_empirical_ starter/results_ model_and_data.md
G9	Not separately measured.	Nothing.	None tested.	Not measured	G9

ID	Measured as	What the data show	Model prediction	Outcome	Evidence
G10	Not measured.	Nothing.	None tested.	Not measured	`G10`
G11	Not measured.	Nothing.	None tested.	Not measured	`G11`
G12	Through the trigger and target of G15, which the data do not show.	No trigger-and-target behaviour (G15).	Identified with λ_S from the thresholds.	Not measured	`K2,06_empirical_starter/results_distress_targeting.md`
G13	As G12.	As G12.	As G12.	Not measured	`K2,06_empirical_starter/results_distress_targeting.md`
G14	Quarters with a dividend (`EQCDIV`), by x.	56.2% of bank-quarters below $y^* = 1.5715$ pay a dividend. Payment falls as x rises, from 60.1% for x between 1.03 and 1.10 to 32.0% between 1.20 and 1.39.	No dividend below y^*.	Fails	`FDIC-Q8,M4,06_empirical_starter/results_dividend_location.md`
G15	Recapitalisations, x before and after, within bank, under the pre-registered rule.	Issue size is unrelated to the pre-issue state and the post-issue state rises with it, $b_{\text{post}} = +1.37$ and $+1.36$ for the two valid distress proxies. Issues start at a mean x of 1.11, against the trigger 1.03.	One target within a bank, $b_{\text{post}} = 0$.	Fails	`FDIC-Q9,FDIC-Q12,M5,06_empirical_starter/PREREG_distress_targeting.md,06_empirical_starter/results_distress_targeting.md`
G16	Not computed. A level would need the economic σ of G8.	Nothing.	v_1^2, v_3^2, which need the cap to bind (M3), and it does not (G2).	Not measured	`M3`
G17	Not computed on this panel.	Nothing.	v_1^2/v_3^2, under the same condition as G16.	Not measured	`K1,M3`
G18	Tier 1 capital over 8.5% of risk-weighted assets.	462 of 130,488 quarter pairs start in deficit, about 0.35%.	None on its own.	Tracked	`FDIC-Q3,FDIC-Q4,06_empirical_starter/results_persistence.md`
G19	Quarterly growth of risk-weighted assets.	Mean -0.47% a quarter on deficit pairs: banks in deficit shrink risk-weighted assets.	Exogenous, so no response to a deficit.	Fails	`FDIC-Q3,M12,06_empirical_starter/results_persistence.md`
G20	One minus dividends over net income, four quarters.	0.3131 on the 108 deficit pairs where it is defined in $(0,1]$.	None on its own.	Tracked	`FDIC-Q7,FDIC-Q8,06_empirical_starter/results_persistence.md`
G21	Share of the log-surplus paid out per quarter.	0.0211 a quarter.	None on its own.	Tracked	`FDIC-Q8,06_empirical_starter/results_persistence.md`
G22	Within the pre-registered tests T1 to T3.	Deficit persistence 0.88 against 1.01 predicted (T1, fails). Surplus persistence 0.9959 against 0.9789 (T3, fails). The fall of ϕ^- in g is not measured (T2).	$\phi^- = 1-g(1-b)/b$, $\phi^+ = 1-\delta_p$.	Fails	`M12,06_empirical_starter/PREREG_persistence.md,06_empirical_starter/results_persistence.md`

5 The ratio constraints in the state

With equity $X - L$, risk-weighted assets πX, high-quality liquid assets $X - a_3\pi X$ and net outflows $a_2 L$, the two ratio constraints of BCVW are the two branches of the cap in $x = X/L$. For $a_1, a_2, a_3, \pi, X, L > 0$,

$$a_1 \le \frac{X-L}{\pi X} \iff \pi x \le \frac{x-1}{a_1}, \qquad 1 \le \frac{X - a_3\pi X}{a_2 L} \iff \pi x \le \frac{x - a_2}{a_3},$$

and both hold if and only if $\pi x \le u(x) = \min\{(x-1)/a_1, (x-a_2)/a_3\}$ (`MeasurementMap.solvency_iff`, `MeasurementMap.lcr_iff`, `MeasurementMap.cap_iff`). At $\pi = 0$ the left side of the first equivalence is undefined and the equivalence fails, and at $a_2 = 0$ the same holds for the second (the two controls). For $\ell_0 < 1$ and $X, L > 0$, $\ell_0 \le (X-L)/X \iff \ell_0/(1-\ell_0) \le x - 1$ (`MeasurementMap.leverage_req_iff`).

6 Pending

- The CET1 ratio (`RBCT1CER`) is not yet in the panel. The corner test of G4 uses Tier 1 against the CET1 minimum (B3-D1).
- The unit of the FDIC amounts, thousands of dollars, is read from one API record and not from the workbook (FDIC-D1).
- Rows G1 to G3 assume regulatory capital equals book equity. Items 12 to 18 and 27 to 29 of Schedule RC-R, which give the adjustments, were not read.
- The tier that a_1 stands for is not fixed (B3-D1).

References

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